

Philosophy in Science

“There’s no such thing as philosophy-free science. There’s only science whose philosophical baggage is taken on board without examination.”

- from *Darwin’s Dangerous Idea* by Daniel Dennett

Duration:	Monday, Jan 5, to Friday, Mar 20	10 weeks total
	Skip Monday, Jan 19th (MLK) & Monday, Feb 16th (President’s Day)	
Time:	2:15 - 3:45 pm, Mondays & Thursdays	1.5 hours twice a week
Location:	Hess Center for Science and Medicine	
	Mondays: usually Hess 5-101 (<i>except for March 2 in Hess 9-121</i>)	
	Thursdays: usually Hess 9-101	
Grade:	Pass/Fail	
Teaching Assistants (TAs):	Cameron Le Roux, B.S.	cameron.leroux@icahn.mssm.edu
	Qi Xiu Fu, B.A.	qixiu.fu@icahn.mssm.edu
	Alessandra Yu, M.Res.	alessandra.yu@icahn.mssm.edu
Director:	Daniela Schiller, Ph.D.	daniela.schiller@mssm.edu

Description

This course dives into the philosophical foundations of science, offering the opportunity to critically reflect on the theoretical underpinning of *your* own research. Science is inherently laden with theory. Theory shapes the questions we ask and how we choose to answer them. Moreover, not all questions are empirical ones. What should we do when two theories equally account for the observed data? While much of science depends on methodological advancement, these theoretical decision points often precede significant scientific breakthroughs and, with enough theoretical pressure, culminate in a paradigm shift.

However, without rigorous theoretical care, uncontrolled proliferation of theories and frameworks without them converging can threaten to overwhelm the scientific landscape of any subfield—fragmenting it and impeding scientific progress. This applies to the life sciences in particular and even more so to biomedical and neural sciences sitting at their intersection, given their multiscale and cognitive aspects. In these burgeoning fields, both our descriptions of *what* we study and our choices for *how* we study our objects of inquiry are both diverse and heavily debated. Even deeper, our motivations for *why* we study what we do and how we choose to study those phenomena are also diverse and rarely disclosed.

We invite you to embark on this theoretical journey through the most nuanced and foundational, yet slippery facets of science. By engaging with theoretical discussion and practical application, we hope you will emerge not only with a better understanding of your place in the field as a scientist, but with enhanced clarity into your own research framework.

- ❖ This course is designed for graduate-level students in the biomedical and neural sciences who are developing foundational thinking for their research interests. We encourage students to actively

discuss and consult with their Principal Investigators (PIs) on the application of the topics and issues raised in this course to their own research, particularly between lecture and practical workshop components in order to get the most out of the course structure. Our aim is to foster a deep interest in the philosophical underpinnings of science, highlighting the inherent role of theory in empirical research as well as the dynamics of scientific progress—and at times, stagnation. By implementing these concepts in concrete exercises made relevant to each student, we hope that theoretical discussion beyond the classroom is enriched and takes further root in the lab, where theory truly meets practice.

- ❖ This course is both an experimental and an evolving process. We value iteration and continuous improvement, particularly as we consider offering this course in future years. Please do provide feedback or share your thoughts on your experiences in integrating the course content with your research.

Objectives

- Identify and describe theoretical **debates** in your own subfield
- Reflect on your **motivations** for the choices you make in your science
- Outline **assumptions** shaping your own research
- Interpret neighboring **subfields** despite the differences in jargon in the literature
- Address **barriers** to theoretical understanding, discussion, and change in everyday science

Expectations

This course consists of **8 modules**, each roughly 1 week long with two 1.5-hour sections.

Every week features a dual structure:

1. **Mondays** are usually **theoretical lectures**, occasionally with guest instructors, paired with heavy discussion. [These guests are well-established researchers themselves, joining us from beyond Sinai, whose topic of research is the theoretical focus of the module.]
2. **Thursdays** are usually **practical workshops** focused on applying the principles of theoretical lecture to your own research with an opportunity to share with classmates and participating PIs.

Active participation is crucial for both segments throughout the course as both are designed to be highly interactive. This course aims to empower each student to build mastery in the theoretical nuances of their own research. It is up to every student to truly apply the principles learned in the theoretical portion not only in the practical workshops, but in their own lab research and broader scientific careers.

Requirements

- **Attendance and participation** are required (**40%**).
 - Students are allowed up to one excused absence.
 - A handheld electronic device must be brought to class for interactive polls and exercises.
- All **readings** are encouraged prior to each session. Extra readings are italicized.
- All **exercises** must be completed prior to the end of the course.
 - This [template](#) can be used for all exercises (see [Project](#) below).
- The **project**, consisting of exercises (**30%**) and a reflection (**10%**), must be submitted at the end.
- All **feedback** forms must be completed following each week (**20%**).

Project

Deadline: Friday, March 20, 2026

The final project includes 2 components:

1. All **exercises**, including a table outlining your theoretical assumptions for your research.
2. A 1-page **reflection** on your experience in this course and how it impacts your research going forward.

Submission Guidelines:

- Copy this [template](#) [File → Make a copy] and replace ‘Template’ with your full name
- Upload into this [folder](#) by deadline

Schedule

	Module	Date	Location	Type	Instructor	Topic
0	Introduction	Mon, Jan 5	Hess 5-101	Lecture	Alessandra	Philosophy in Science
		Thu, Jan 8	Hess 9-101	Workshop	Daniela Schiller	Framework Table
1	Context	Mon, Jan 12	Hess 5-101	Lecture	Cameron	Pragmatic Approach
		Thu, Jan 15	Hess 9-101	Guest Lecture	Carla Sharp	Alternative DSM
		Mon, Jan 19	Hess 5-101	MLK Day	-	-
		Thu, Jan 22	Hess 9-101	Workshop	Cameron	Background Context
2	Virtues	Mon, Jan 26	Hess 5-101	Guest Lecture	Milena Ivanova	Aesthetics in Science
		Thu, Jan 29	Hess 9-101	Workshop	Daniela Schiller	Systematizing Virtues
3	Phenomenon	Mon, Feb 2	Hess 5-101	Guest Lecture	Paul Linton	Visual Phenomena
		Thu, Feb 5	Hess 9-101	Workshop	Cameron	Phenomenon Map
4	Mechanism	Mon, Feb 9	Hess 5-101	Guest Lecture	Lauren Ross	Scientific Mechanisms
		Thu, Feb 12	Hess 9-101	Workshop	Qi Xiu	Mechanism Map
		Mon, Feb 16	Hess 5-101	President's Day	-	-
-	Paradigm Shifts	Thu, Feb 19	Hess 9-101	Guest Lecture	Charles Mobbs	Paradigm Shifts
5	Purpose	Mon, Feb 23	Hess 5-101	Guest Lecture	Nichole Levesley	Typology of Teleology
		Thu, Feb 26	Hess 9-101	Workshop	Alessandra	Teleology Table
6	Framework	Mon, Mar 2	Hess 9-121	Presentations	Daniela Schiller	-
		Thu, Mar 5	Hess 9-101	Feedback	Cameron	-
7	Beyond	Mon, Mar 9	Hess 5-101	Workshop	Daniela Schiller	Writing by Framework
		Thu, Mar 12	Hess 9-101	Guest Lecture	Georg Northoff	Neurophenomenology
-	Project	Fri, Mar 20	<i>Drive</i>	-	-	-

Syllabus

Module 0 Intro to Philosophy in Science: *What are we even doing?*

In this foundational module, students will be introduced to the goals and expectations of the course. This will begin with an introduction to philosophy *of* science, followed by delving into considerations of philosophy *in* science. Then, we will explore why scientists should care about philosophy and what barriers we face in practicing philosophy in science. Finally, students will be introduced to the template of assumptions that gives the course its structure—a scaffold that we will fill in and build on in each new module with the end result being an outline of each student's theoretical framework for their research. In essence, we introduce the question—*what are we even doing in our science?*—and provide a roadmap for answering it.

Theoretical Lecture Monday, Jan 5, 2026

Hess 5-101

Instructors: Dr. Daniela Schiller & Alessandra Yu

To understand what we're doing, it helps to have an understanding of what science is. In this foundational module, we'll be delving into the core question in philosophy *of* science: what is science? We'll explore the range of topics within this overarching inquiry to get a handle on what's at stake in this question, including theoretical considerations in empirical research: philosophy *in* science. Then, we'll identify potential challenges that scientists face in embracing the full scope of our field and examine how these challenges influence theoretical considerations.

Readings

- ☐ For biomedical students:
 - Kampourakis & Uller (2020) - Philosophy of Biology: [Preface & Chapter 1](#) (pg 1-18)
- ☐ For neuroscience students:
 - Bechtel & Huang (2022) - Philosophy of Neuroscience: [Introduction](#) (pg 1-2)
 - Levenstein et al. (2023) - [On the Role of Theory and Modeling in Neuroscience](#)
- ☐ Schiller, Yu, et al. (2024) - [The Human Affectome: Intro](#) (example of philosophy in science)

Exercise

- ☐ Fill out a [table](#) with two columns on barriers to practicing philosophy in science:
 - On the left, create a ranked list of barriers that scientists face when getting into theoretical discussion about their research.
 - On the right, describe the barriers that you face with regards to theoretical issues in *your* particular research or in your corner of your subfield.

Practical Workshop Thursday, Jan 8, 2026

Hess 9-101

For this initial practical workshop of the course, students begin the essential process of mapping out their research through a structured Table of Assumptions. This exercise is designed to help each participant explicitly articulate the foundational assumptions that underpin their scientific investigations. Students can take their first stab at the table and share it during the workshop. Each subsequent module will elaborate on a domain in the table, so we will keep coming back to it until the end of the course. This

table summarizes domains of theoretical assumptions for their scientific framework which shapes the scientific theory each student uses.

Exercise

- ☐ **Table of Assumptions:** familiarize yourself with and fill out a [table](#) template of assumptions shaping your research:
 - **Context:** what is the background for studying what you do the way you do?
 - **Virtues:** what makes your theory good?
 - **Phenomenon:** what are you trying to study?
 - **Construct:** what are you studying practically?
 - **Mechanism:** how does what you're studying work?
 - **Operationalization:** how are you trying to study it practically?
 - **Purpose:** why do you think your phenomenon exists?

Feedback

- ☐ Please provide your feedback on **Module 0:** Intro to Philosophy in Science [here](#). This will help us refine the course to better meet students' needs and expectations.

Module 1 Context: *Why study this phenomenon this way?*

This lecture dives into the essential *why*-questions that underpin all scientific inquiry, specifically examining the motivations behind studying certain phenomena in science. We will explore how these motivations are not just personal choices but are often deeply influenced by the broader context in which the research occurs. This module will help students recognize the complex interplay between individual interests and the historical, cultural, and academic contexts that shape these interests.

Theoretical Lecture

Monday, Jan 12, 2026

Hess 5-101

Instructors: Cameron Le Roux & Dr. Daniela Schiller

In this lecture, we will explore the motivations driving scientific inquiry and how these motivations shape research methods and outcomes. We will discuss the pragmatic aspects of science, focusing on how the context of research affects the questions asked and the explanations provided. The lecture will highlight the role of philosophy in understanding scientific motivation and pragmatism, delving into the *why*-questions that underpin research. We will also examine biases stemming from local research traditions, practical observation methods, and audience interests. By the end, students will gain insights into the philosophical and pragmatic challenges influencing their research.

Readings

- ☐ Van Fraassen (1980) - The Scientific Image: [Chapter 6 - Pragmatic Theory](#) (pg. 136-137)
- ☐ Smaldino & McElreath (2016) - [The natural selection of bad science](#)

Guest Lecture

Thursday, Jan 15, 2026

Hess 9-101

Guest Lecturer: Dr. Carla Sharp

Dr. Carla Sharp is a world-renowned researcher in the field of personality pathology. She is currently a John and Rebecca Moores Professor and Associate Dean for Faculty and Research in the College of Liberal Arts and Social Sciences at the University of Houston. She is also Director of the Adolescent Diagnosis Assessment Prevention and Treatment Center and the Developmental Psychopathology Lab at UH. She has a longstanding interest in social cognition (mentalizing) as a cause and correlate of psychiatric disorder across the lifespan with a special focus on youth. Her lecture will cover her contributions and those of her colleagues to a paradigm shift in the “context” of personality disorder research and treatment: the introduction and acceptance of dimensional models of personality disorders, and ongoing efforts to revise the Diagnostic and Statistical Manual, 5th Edition (DSM-5).

Readings

- ☐ Sharp et al. (2025) - [The validity, reliability and clinical utility of the Alternative DSM-5 Model for Personality Disorders \(AMPD\) according to DSM-5 revision criteria](#)
- ☐ Sharp & Wall (2021) - [DSM-5 Level of Personality Functioning: Refocusing Personality Disorder on What It Means to Be Human](#)
- ☐ *Optional:*
 - ☐ Sharp & Miller (2024) - [Head-to-Head Comparisons of Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition, Section II and III Personality Disorder in Predicting Clinical Outcomes](#)
 - ☐ Sharp & Miller (2022) - [Ten-Year Retrospective on the DSM-5 Alternative Model of Personality Disorder: Seeing the Forest for the Trees](#)

No class on Monday, Jan 19, 2026 - Martin Luther King Jr. Day

Practical Workshop

Thursday, Jan 22, 2026

Hess 9-101

In this practical workshop, students will apply the concepts discussed in the theoretical lecture to their own research motivations. Through the completion of a reflective exercise, students will explore the underlying reasons for their research focus and how historical and contextual factors have shaped these motivations. During the workshop, students will share their completed exercises to spur discussion. This session is designed to deepen students' understanding of the philosophical and pragmatic drivers of their scientific inquiries, fostering a more introspective and comprehensive approach to their work.

Exercise

- ☐ Track the history of implicit or explicit motivations in your subfield and reflect. (Ask yourself: *what has led you, your lab, and your field to ask the scientific question you are asking?*)
- ☐ (Optional) Trace your academic lineage while highlighting the contextual motivations influencing each researcher that came before you. (e.g., use [Neurotree](#) to see your neuroscience family tree)

Feedback

☐ Please provide your feedback on **Module 1**: Context [here](#).

Module 2 Virtues: *What makes your theory good?*

This week, we focus on the theoretical virtues that determine the robustness and quality of scientific theories. Students will explore how empirical data often fail to resolve disputes between competing theoretical frameworks, necessitating an examination of the underlying virtues that guide theory selection. By the end of the module, students will understand the key qualities that make a theory scientifically valuable and how to apply these criteria to their own research.

Theoretical Lecture *Monday, Jan 26, 2026*

Hess 5-101

Instructors: Alessandra Yu & Dr. Daniela Schiller

In this lecture, we will explore the concept of *theoretical virtues*—features that define a good scientific theory. We will examine classic and contemporary perspectives on these virtues, such as simplicity, explanatory power, and coherence, and discuss how they help arbitrate between competing theories. The lecture will include discussions on Ockham’s Razor and Sober’s interpretations of different kinds of simplicity. By the end of the session, students will gain insights into how these virtues influence scientific debates and theory development.

Guest Lecturer: Dr. Milena Ivanova

Dr. Milena Ivanova is a philosopher of science whose work focuses on the aesthetics of science, scientific creativity, and feminist perspectives on science and technology. She is a Senior Teaching Associate at the Leverhulme Centre for the Future of Intelligence, a Course Director in Philosophy at the University of Cambridge Institute of Continuing Education, and a Bye-Fellow at Newnham College, Cambridge. Her work explores the aesthetic, epistemic, and ethical dimensions of scientific practice, including why science is often described as beautiful, how non-empirical values guide theory choice, and in what ways experiments can be appreciated aesthetically. She is the author of *Duhem and Holism* (Cambridge University Press, 2021) and co-editor of *The Aesthetics of Science: Beauty, Imagination and Understanding* and *The Aesthetics of Scientific Experiments*. Her work has been recognized with multiple awards, including the Routledge, Taylor & Francis Prize for “*What Is a Beautiful Experiment?*” Dr. Ivanova is also a governing committee member of the British Society for Philosophy of Science, member of the British Society for Aesthetics, and the American Philosophical Association.

Reading

☐ Ivanova & Farr (2020) - [Methods in Science and Metaphysics](#)

Practical Workshop *Thursday, Jan 29, 2026*

Hess 9-101

For this workshop, students will apply their understanding of theoretical virtues to real-world scientific debates. By completing the exercise below for the workshop, students can learn to critically assess the strengths and weaknesses of different theoretical frameworks and foster a deeper appreciation for the role of virtues in scientific inquiry, with the content of each student’s exercise driving workshop discussion.

Reading

- ☐ Keas (2017) - [Systematizing the Theoretical Virtues](#)

Exercise

- ☐ Describe the underlying theoretical argument in a debate in your field (in light of theoretical virtues):
 - Compare and contrast the theoretical sides in the debate in a table.

Feedback

- ☐ Please provide your feedback on **Module 2: Virtues** [here](#).

Module 3 Phenomenon: *What are you trying to study?*

This week, we delve into the nature of scientific phenomena and the metaphysical assumptions that underpin them. As scientific communities grow, divergent assumptions about what constitutes a phenomenon can lead to theoretical divisions. Students will explore how different academic communities define and approach phenomena, shaping their research questions and methodologies. By the end of the module, students will have a clearer understanding of how to identify and frame the phenomena they study.

Theoretical Lecture

Monday, Feb 2, 2026

Hess 5-101

Instructors: Cameron Le Roux & Dr. Daniela Schiller

In this lecture, we will examine the philosophical concept of phenomena and how different assumptions about their nature can influence scientific theories. We will discuss the metaphysical foundations of scientific phenomena, focusing on how different academic communities define and study these constructs. The lecture will explore key philosophical literature on scientific explanation and the abstraction of phenomena, providing students with a framework to understand and articulate the phenomena central to their research. Additionally, we will highlight an example of a phenomenon where philosophical considerations significantly altered scientific understanding through the insights of our guest speaker, Dr. Paul Linton.

Guest Lecturer: Dr. Paul Linton

Dr. Paul Linton is a neuroscientist and philosopher specializing in 3D vision. He received his PhD in 2021 from the Center for Applied Vision Research, City, University of London, where his research challenged our understanding of distance perception by showing the visual system is unable to triangulate distance using the two eyes. He was also part of the DeepFocus team at Meta Reality Labs. Paul is the author of *The Perception and Cognition of Visual Space* (Palgrave Macmillan, 2017). Before vision science, he was a stipendiary lecturer in law at St Hilda's College, Oxford University, and a teaching fellow in philosophy at University College London. As a Presidential Scholar, Paul will develop his new two-stage theory of 3D vision using the latest techniques in machine learning and fMRI in the hope of explaining how we experience the 3D world.

Readings

- ☐ Bogen & Woodward (1988) - [Saving the Phenomena](#)

- ☐ SEP (2025) - [20th Century Theories of Scientific Explanation](#) (Intro, 1, 2.1, 3.1, 4.1, 5.1, & 7)
- ☐ Firestone & Scholl (2015) - [Cognition does not affect perception: Evaluating the evidence for “top-down” effects](#)
- ☐ Oliver Sacks (2006, *New Yorker*) - [Stereo Sue](#)

Practical Workshop

Thursday, Feb 5, 2026

Hess 9-101

In this workshop, students will apply the concepts from the theoretical lecture to their own research. Prior to the workshop, students are expected to complete one of the exercises below in order to guide and drive the discussion during the workshop, helping students critically assess how different constructions of phenomena influence scientific inquiry.

Exercise

- ☐ Draw a conceptual map of:
 - Your phenomenon and related phenomena (i.e., your ontology)
 - Your construct and related constructs (i.e., your conceptual map)
 - **Your phenomenon you’re trying to study related to the constructs you’re studying**
- ☐ Consider: *what neighboring subfields have constructs that refer to the same phenomenon or overlap with yours?*

Feedback

- ☐ Please provide your feedback on **Module 3: Phenomenon** [here](#).

Module 4

Mechanism: How does the phenomenon work?

This week, we explore the mechanisms underlying scientific phenomena, focusing on how different assumptions about these mechanisms shape scientific theories. Students will examine how scientific phenomena arise and the methods used to observe and measure these processes. The module will also address the challenges of studying living phenomena, particularly in the context of emergence and complexity. By the end of the week, students will have a deeper understanding of how to operationalize and investigate the mechanisms underlying their research topics.

Theoretical Lecture

Monday, Feb 9, 2026

Hess 5-101

Instructors: Qi Xiu Fu & Dr. Daniela Schiller

In this lecture, we will delve into the concept of mechanisms in science, exploring how different theoretical assumptions about these mechanisms influence scientific explanations. We will discuss classic and contemporary views on the mechanisms of phenomena, examining how they arise, function, and can be observed and measured. Special emphasis will be placed on the unique challenges posed by living phenomena, such as the concept of emergence and the debate over whether complexity alone is sufficient to account for life. We will explore how these ideas impact the operationalization of scientific theories and the methodologies employed in empirical research. By understanding these underlying mechanisms, students will be better equipped to develop and refine their scientific investigations.

Guest Lecturer: Dr. Lauren Ross

Dr. Lauren Ross is Dean's Professor and Chancellor's Fellow at the Logic and Philosophy of Science Department at the University of California, Irvine. She received my Ph.D. from the History and Philosophy of Science Department at the University of Pittsburgh, her M.A. in bioethics from the Center for Bioethics and Health Law at the University of Pittsburgh, and her M.D. from the University of California, Irvine. Supported by a National Science Foundation (NSF) CAREER award, John Templeton Foundation Grant Awards, and a Humboldt Experienced Researcher Fellowship, sponsored by the Munich Center for Mathematical Philosophy (MCMP) at Ludwig Maximilian University of Munich (LMU), her research concerns explanation and causation in biology, neuroscience, and medicine. Her work involves interrelated projects that address: the nature of explanation in these sciences, different causal structures and explanation types, and the rationale that guides particular forms of causal reasoning in these domains.

Readings

- ☐ SEP (2019) - [Mechanisms in Science](#)
- ☐ For neuroscientists:
 - Ross & Bassett (2024) - [Causation in neuroscience: keeping mechanism meaningful](#)
- ☐ Bassett & Ross (*The Transmitter*, 2024) - [What are mechanisms? Unpacking the term is key to progress in neuroscience](#)

Practical Workshop *Thursday, Feb 12, 2026*

Hess 9-101

In this workshop, students will apply the theoretical concepts discussed earlier in the week to their own research. Students will complete an exercise designed to analyze the mechanisms in their subfield or research. During the session, they will use their reflections to foster discussion and gain deeper insights into operationalizing these mechanisms. This hands-on approach will help students critically evaluate different theoretical perspectives and develop robust research strategies.

Exercise

- ☐ Transform your phenomenon or construct map into a mechanism and/or operationalization map.
- ☐ *Design a study to parse between the conflicting theoretical sides in your subfield's debate.*

Feedback

- ☐ Please provide your feedback on **Module 4: Mechanism** [here](#).

No class on Monday, February 16, 2026 - President's Day

Mid-Course Lecture on Paradigm Shifts

Theoretical Lecture *Monday, Feb 19, 2026*

Hess 9-101

Instructor: Dr. Daniela Schiller

In this lecture, we will examine Thomas Kuhn's account of scientific revolutions and paradigm shifts, focusing on how periods of "normal science" give way to conceptual rupture when anomalies accumulate and existing frameworks can no longer accommodate them. We will explore what paradigms are, why

paradigm change is not simply a matter of adding new data, and how shifts in standards, methods, and values reshape what scientists take the world to be like. The lecture will also consider the implications of Kuhn's view for scientific progress, objectivity, and disagreement across theories.

Guest Lecturer: Dr. Charles Mobbs

Dr. Charles Mobbs is a Professor of Neuroscience, Endocrinology, and Geriatrics at the Icahn School of Medicine at Mount Sinai. His research investigates the biological mechanisms linking metabolism, aging, and age-related disease, with a particular focus on dietary restriction, glucose signaling, innate immunity, and neurodegenerative disorders such as Alzheimer's disease. He received his BS in Biology from MIT and his PhD in Cell and Molecular Biology from the University of Southern California, followed by NIH-supported postdoctoral training at Rockefeller University. In addition to his academic work, he is the Founder and CEO of Gilga-Med Inc., a translational research company developing therapies for diseases of aging. Across academia and industry, his work exemplifies how scientific discovery is shaped by conceptual reframing, attention to underlying assumptions, and careful engagement with data.

Reading

- ☐ Kuhn (1977) - [Second Thoughts on Paradigms](#)

Feedback

- ☐ Please provide your feedback on **Paradigm Shifts** [here](#).

Module 5 Purpose: *Why does the phenomenon exist?*

This week, we delve into the teleological aspects of scientific phenomena, examining the underlying purposes behind their existence. Understanding the purpose of phenomena is crucial for integrating and comparing different theoretical frameworks. Students will explore how teleological principles help bridge metaphysical and mechanistic assumptions, providing a cohesive framework for scientific inquiry. By the end of the module, students will be able to articulate the purpose of their research phenomena and evaluate theories in light of this understanding.

Theoretical Lecture

Monday, Feb 23, 2026

Hess 5-101

Instructors: Alessandra Yu & Dr. Daniela Schiller

In this virtual lecture, we will explore the concept of purpose in scientific phenomena, distinguishing between extrinsic and intrinsic purposes. We will discuss how understanding the purpose of phenomena can help bridge gaps between different metaphysical and mechanistic theories. The lecture will highlight how a teleological framework can organize and evaluate theories, integrating scholarly and theoretical values to achieve a comprehensive understanding of phenomena. Additionally, we will examine the role of algorithms in identifying and modeling the purposes of phenomena, discussing how computational approaches can enhance our understanding of teleological principles. Students will learn to distinguish purpose from function and goal and understand the significance of teleological assumptions in scientific research.

Guest Lecturer: Nichole Levesley

Nichole Levesley is a PhD student in the Department of History and Philosophy of Science at the University of Cambridge. Her research focuses on the history and philosophy of astronomy and cosmology, with particular attention to how limits on scientific knowledge have been conceptualized in modern astrophysics and cosmology. She is currently investigating the emergence of infrared astronomy in the 1960s–1980s and has broader interests in historiography and integrated approaches to HPS. She also organizes a Science Fiction and History & Philosophy of Science reading group, which explores the role of speculative imagination in scientific and philosophical inquiry.

Readings

- ☐ Levesley (2025) - [A Classification in Teleology in Biology & Cosmology](#)
- ☐ SEP (2020) - [Teleological Notions in Biology](#)
- ☐ SEP (2024) - [Levels of Organization in Biology](#)
- ☐ For neuroscientists (optional):
 - SEP (2020) - [Emergent Properties](#)
- ☐ For biomedical scientists (optional):
 - Dennis Noble (2024 book review) - [It's time to admit that genes are not the blueprints for life](#)

Practical Workshop

Thursday, Feb 26, 2026

Hess 9-101

In this workshop, students will reflect on the purpose of their research phenomena, considering whether the purpose is intrinsic or extrinsic. Prior to the workshop, students are expected to complete the exercise, which involves critically evaluating the teleological aspects of their phenomena. During the session, students will share their reflections, and guiding discussions on how understanding purpose can bridge their metaphysical and mechanistic assumptions. This collaborative exploration will deepen students' appreciation of the role of teleology in scientific inquiry.

Exercise

- ☐ Use the provided [table](#) to reflect on the purpose of your phenomenon, through each of the possible classificatory aspects, and consider whether it is intrinsic or extrinsic.

Feedback

- ☐ Please provide your feedback on **Module 5**: Purpose [here](#).

Module 6

Framework: How do we choose to do science?

This module examines the underlying frameworks guiding scientific inquiry, focusing on how implicit assumptions influence every stage of the scientific process, from experimental design and methodology to analysis and interpretation. Students will reflect on and modify their research frameworks based on insights gained throughout the course, culminating in presentations that showcase their refined approaches to scientific inquiry, split between Monday and Thursday workshop sections.

Exercise

- ☐ Return to the framework table you created in Module 1 and modify based on what's come out of the recent workshops.

Practical Workshop Presentations *Monday, Mar 2, 2026* **Hess 9-121**

In this workshop, students will revisit the framework table they created in Module 1, incorporating new insights and modifications based on the recent workshops. This session will involve presenting and discussing these updated frameworks, allowing students to receive constructive feedback and refine their approaches further. The goal is to ensure that each student's framework is comprehensive and well-grounded in the theoretical and practical concepts covered in the course. By the end of this session, students will have a solidified understanding of their scientific frameworks and how to enhance them.

Exercise

- ☐ Present your modified framework table, demonstrating how your understanding and approach to scientific inquiry have evolved over the course. This comprehensive presentation will highlight the integration of theoretical and practical knowledge, showcasing your ability to critically evaluate and enhance your scientific frameworks.

Feedback (optional)

- ☐ Please provide your feedback on **Module 6: Framework** [here](#).

Practical Workshop Feedback *Thursday, Mar 5, 2026* **Hess 9-101**

The second workshop session will conclude the main portion of the course allowing reflection on students' experience in developing and refining their frameworks. All feedback will be pivotal for future iterations of the course.

Feedback

- ☐ Please provide your feedback on the **entire course** based on your experience [here](#) for future iterations.

Module 7 *Beyond: How else can we do science?*

In this final module, we will explore alternative conceptualizations of science and consider new directions for scientific inquiry. While this course has focused on certain philosophical positions, it is important to acknowledge that there are diverse perspectives on what science is and how it can be conducted. Students will be introduced to various approaches, including neurophenomenology and the role of subjectivity in science, and will discuss ways to overcome barriers to theoretical discussion and change within neuroscience. This module encourages students to think beyond traditional frameworks and embrace innovative approaches to scientific research, fostering a more inclusive and dynamic scientific community.

Practical Workshop *Monday, Mar 9, 2026*

Hess 5-101

In this final workshop, students will reflect on how their revised scientific frameworks influence their writing and communication strategies. During the workshop, students will complete an exercise that involves orienting a reader in their scientific writing, whether in the first paragraph of their own work or a review. During the session, students will share their approaches and discuss how their frameworks guide the way they present and communicate their research. Additionally, this session will gather valuable feedback from students about their course experience, aiming to improve future iterations of the course.

Suggested Reading

- ☐ Matthews (2014) - Successful Scientific Writing - [Chapter 1: Start with a Plan](#)

Exercise

- ☐ Write the first paragraph of your own review and discuss: how would you orient a reader?

Theoretical Lecture *Thursday, Mar 12, 2026*

Hess 9-101

Instructors: Cameron Le Roux & Dr. Daniela Schiller

In this lecture, students will be introduced to alternative philosophical perspectives on science that challenge conventional methodologies. We will consider the concept of science being atheoretical, where empirical data is prioritized over theoretical frameworks. The lecture will also cover neurophenomenology, which integrates subjective experiences with objective neuroscience to provide a holistic understanding of the mind and brain. Additionally, we will explore the role of the self in scientific inquiry, questioning how perception and agency can be incorporated into our science and influence research outcomes. Students will be encouraged to critically evaluate these perspectives and consider how they might integrate these alternative approaches into their own scientific practice. Finally, Dr. Georg Northoff will illustrate how to incorporate neurophenomenology into scientific understanding and research methodologies:

Guest Lecturer: Dr. Georg Northoff

Dr. Georg Northoff is a psychiatrist, neuroscientist, and philosopher whose work investigates the brain–mind relationship underlying self, consciousness, and subjectivity. He is a Professor and Senior Scientist affiliated with Carleton University and serves as Director of Mind, Brain Imaging, and Neuroethics at The Royal's Institute of Mental Health Research in Ottawa. His research combines brain imaging, philosophy, and clinical psychiatry to examine how spatiotemporal brain dynamics give rise to mental phenomena, with implications for understanding and treating psychiatric disorders such as depression, schizophrenia, anxiety, and bipolar disorder. He is the author of several influential books, including *The Spontaneous Brain*, *Neurowaves: Brain, Time, and Consciousness*, and *From Brain Dynamics to the Mind: Spatiotemporal Neuroscience*.

Reading

- ☐ Northoff & Ventura (2025) - [Bridging the gap of brain and experience: Converging Neurophenomenology with Spatiotemporal Neuroscience](#)
- ☐ Ward (2017) - [The Varieties of Enactivism](#)

Exercise

- ☐ Suggest ways that the fields of biomedical and neural sciences can break down the barriers for theoretical discussion, pushback, and change.

Feedback

- ☐ Please provide your feedback on **Module 7**: Beyond [here](#).

Project Due *Friday, March 20, 2026* See full project description above.

Please **submit** your Google Doc with the items below in this [folder](#) by the deadline:

- ☐ All **exercises** from Module 0-7
- ☐ A 1-page **reflection** on your experience in the course and how it impacts your scientific research and your approach to it moving forward